Practice Sheet – Logic & Binary Systems

Week 2 | Introduction to Computer Science

Submission Details: This practice sheet must be submitted via iLearn. Please refer to your lecture notes for the complete submission process and required document format.

Task 1: Logic Simplification, Equivalence, and Validity

Part (a): Simplify the Expressions

Using logical equivalences (such as De Morgan's Laws, Identity Laws, and Absorption Laws), simplify each of the following:

1. $\neg(P \vee Q)$
2. $(P \wedge True) \vee (P \wedge False)$
3. $(P \vee Q) \wedge (P \vee \neg Q)$

Show all working steps and state which law you apply at each stage.

Part (b): Validity Analysis

Determine whether each expression is a **tautology** (always true), a **contradiction** (always false), or **satisfiable** (sometimes true):

1. $P \vee \neg P$
2. $P \wedge \neg P$
3. $(P \Rightarrow Q) \vee (Q \Rightarrow P)$

Justify your answers using truth tables or logical reasoning.

Task 2: Truth Tables

Part (a): Construct Truth Tables

Build complete truth tables for the following expressions. Include columns for all intermediate steps:

1. $P \wedge (Q \vee R)$
2. $(P \vee Q) \Rightarrow R$
3. $(P \oplus Q) \oplus R$ where $\oplus$ denotes XOR (exclusive OR)

Remember: XOR is true when inputs differ, false when they match.

Part (b): Equivalence Testing

Use your truth tables from standard equivalences to verify whether these pairs are logically equivalent:

1. $\neg(P \vee Q)$ and $\neg P \wedge \neg Q$
2. $P \Rightarrow Q$ and $\neg P \vee Q$

Two expressions are equivalent if their truth values match in every possible case.

Task 3: Boolean Equations, Symbol Notation, and Logic Gates

Part (a): Boolean Equations

Write the Boolean equation for:

- AND gate (inputs A, B)
- OR gate (inputs A, B)
- NAND gate (inputs A, B)

Use standard notation: ∧ for AND, ∨ for OR, ¬ for NOT.

Part (b): Gate Symbols

Draw the standard symbolic notation for each of the three gates listed in part (a).

Label inputs (A, B) and outputs clearly.

Part (c): Truth Tables

Complete truth tables showing all input combinations and corresponding outputs for:

- AND gate
- OR gate
- NAND gate

Task 4: Binary Numbers and Arithmetic

01

Decimal to Binary Conversion

Convert these decimal numbers to 8-bit binary (unsigned representation):

- 45
- 200

02

Two's Complement Representation

Represent these negative numbers in 8-bit Two's Complement format:

- −18
- −73

Hint: Convert the positive value to binary, invert all bits, then add 1.

03

Binary Addition

Perform these binary additions (show carries):

- $01011010_2 + 00010101_2$
- $11101001_2 + 11110011_2$ (interpret as Two's Complement)

04

Binary Subtraction Using Two's Complement

Perform these subtractions in 8-bit binary using Two's Complement method:

- *37 - 25*
- *23 - 89*

Remember: A − B = A + (−B) in Two's Complement.

Assessment Guidance

This practice sheet covers fundamental concepts in logic and binary systems essential for computer science. Take your time with each task, show all working steps, and verify your answers. These skills form the foundation for understanding digital circuits, computer architecture, and algorithm design. If you encounter difficulties, review your lecture notes.

Reminder: Submit your completed work via iLearn. Check the lecture notes for formatting requirements and the exact submission deadline. Good luck!